A Rate–Distortion View of Semantic Information

Numerical Reproduction of Binary and Gaussian Case Studies

EE142 Information Theory Course Project | ShanghaiTech University

Semantic information is modeled as reconstructing both meaning and appearance

The paper represents an information source by a hidden semantic state S and an observable appearance X. The encoder only observes X, but the decoder must reproduce both $\hat{S}$ and $\hat{X}$.

$$R(D_s, D_a) = \min I(X; \hat{S}, \hat{X})$$

subject to semantic distortion D_s and appearance distortion D_a.

- S: intrinsic state or semantic content.
- X: extrinsic observation or appearance.
- D_s: task/meaning distortion.
- D_a: signal/appearance distortion.

Reproduction strategy turns theorem consequences into tests

- Boundary conditions become unit tests.
- Paper formulas become deterministic numerical routines.
- Figures are regenerated from tracked scripts and CSV files.

Track	Coverage
Binary classification	Prop. 5 and Prop. 6
Gaussian source	Prop. 3 and Prop. 4
Verification	26 automated tests

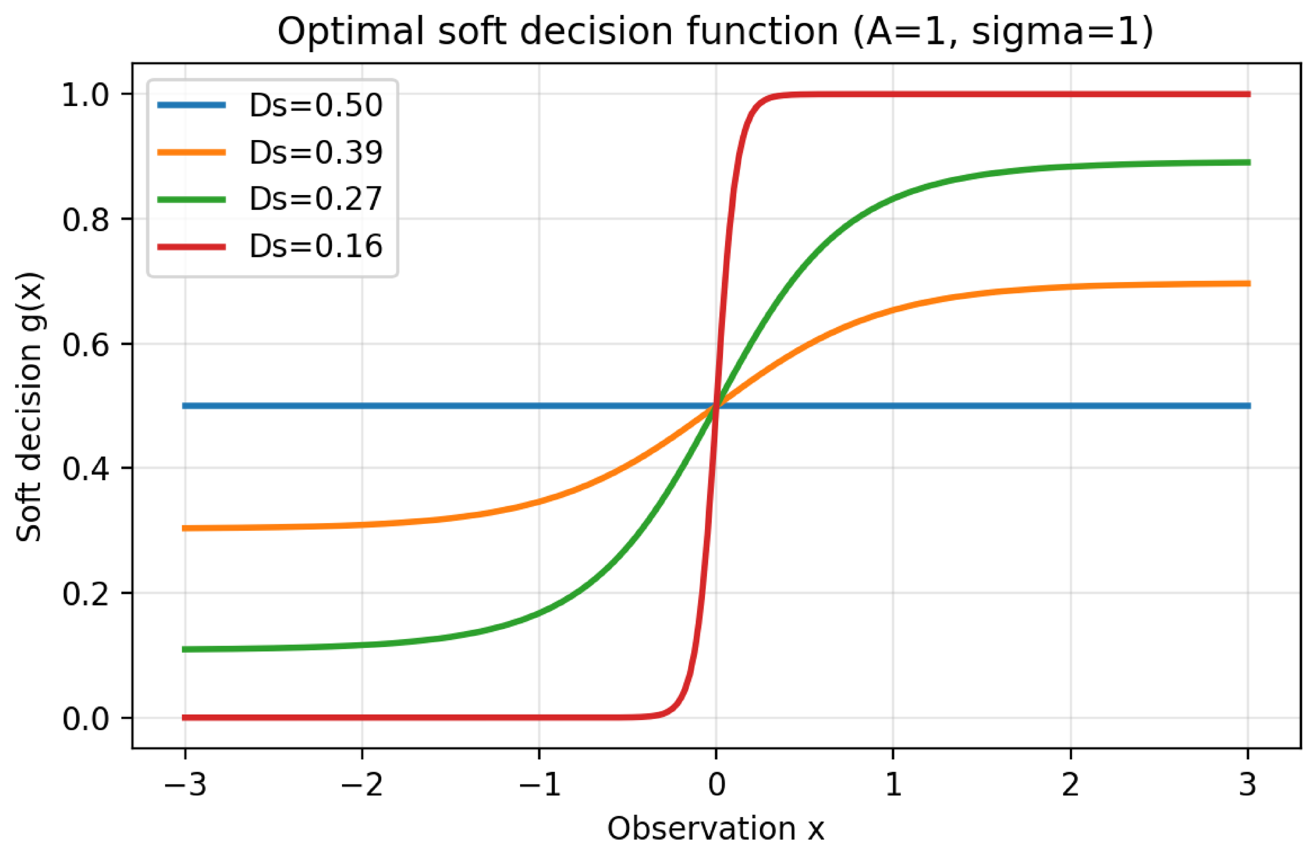
Binary classification gives a clean semantic distortion boundary

For $S \in \{0, 1\}$ and

$$X|S = 0 \sim \mathcal{N}(A, \sigma^2), \quad X|S = 1 \sim \mathcal{N}(-A, \sigma^2),$$

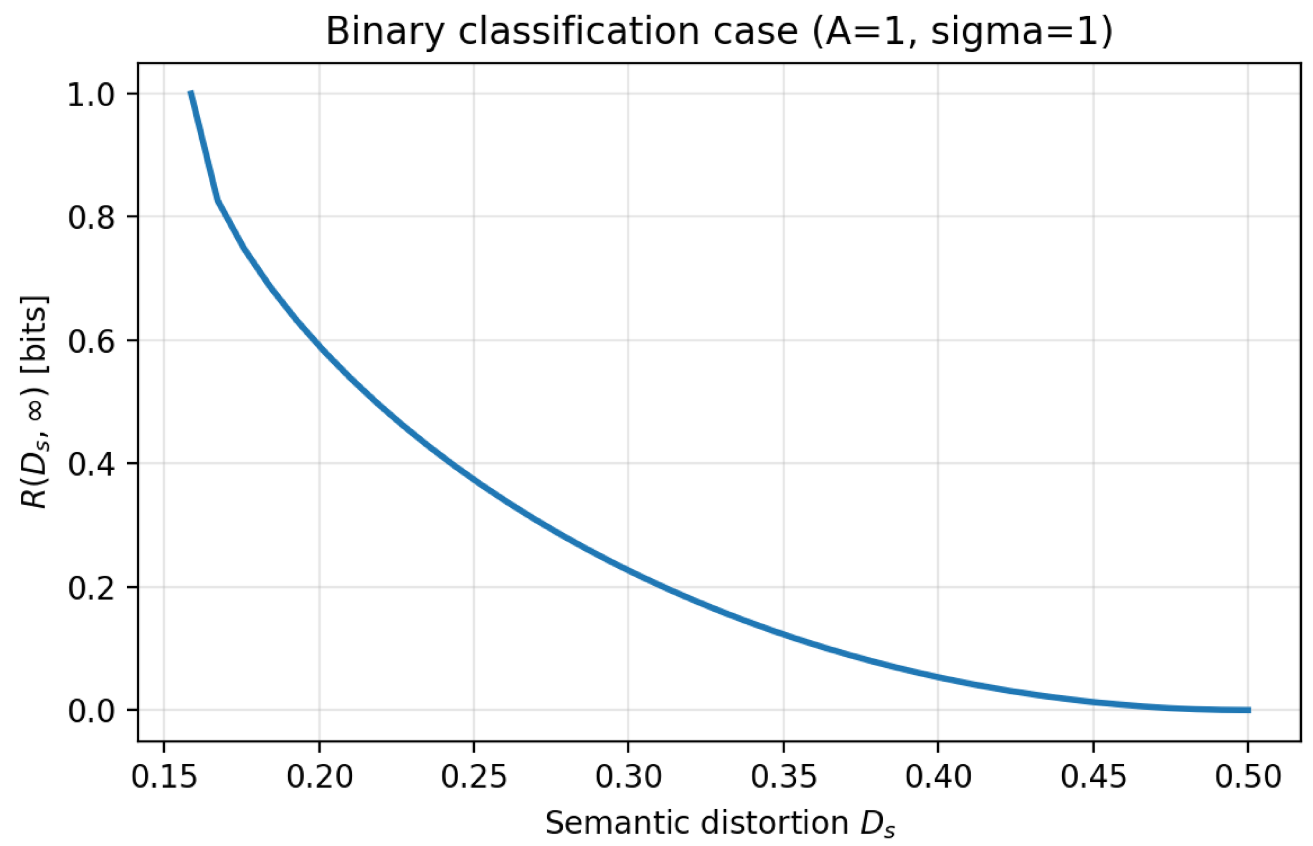
the minimum feasible semantic distortion is the Bayes error $Q(A/\sigma)$.

The optimal soft decision changes continuously with the allowed semantic distortion



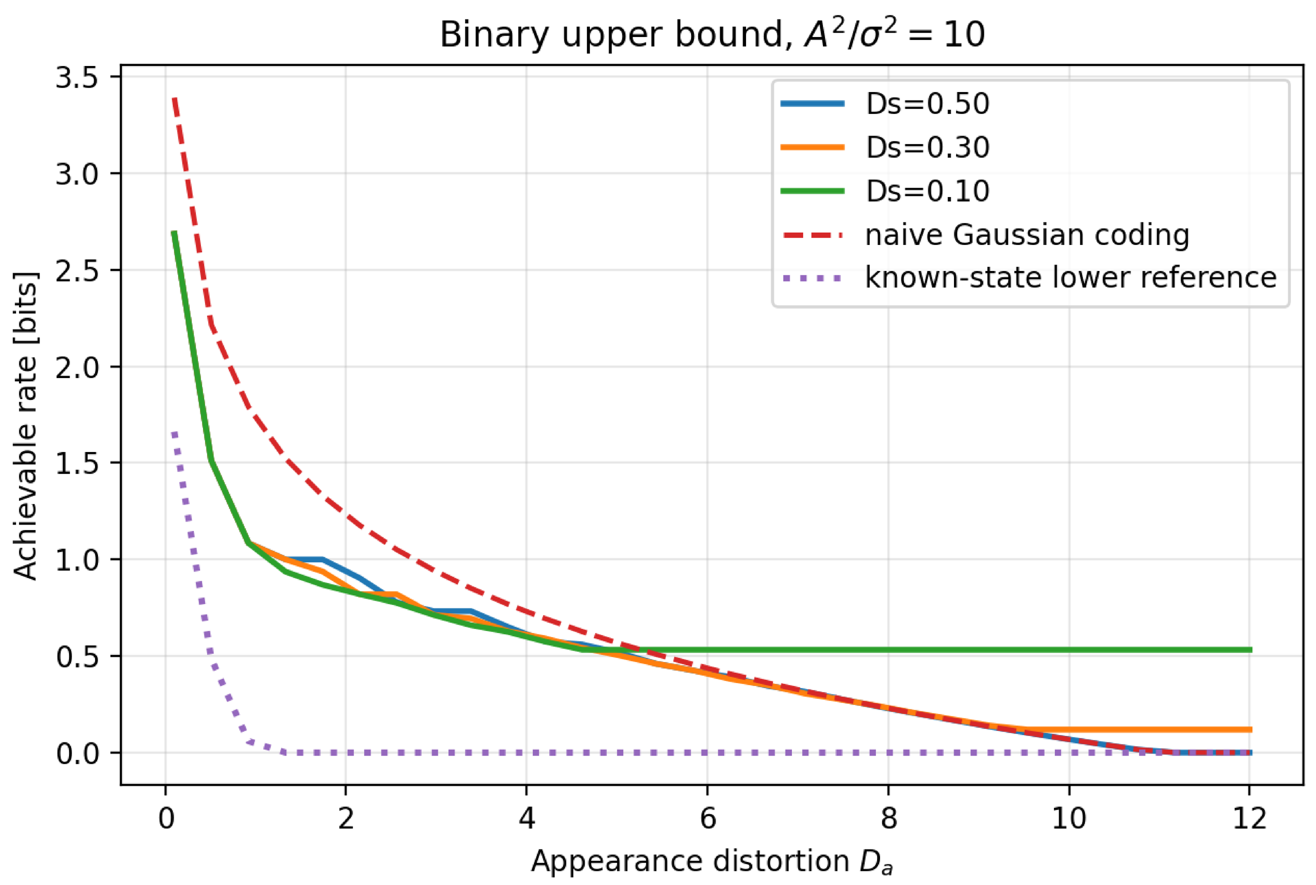
The soft decision function $g(x) = P(\hat{S} = 0|x)$ becomes sharper when D_s approaches the Bayes error, and flatter as larger semantic distortion is allowed.

The binary semantic rate decreases as semantic distortion is relaxed



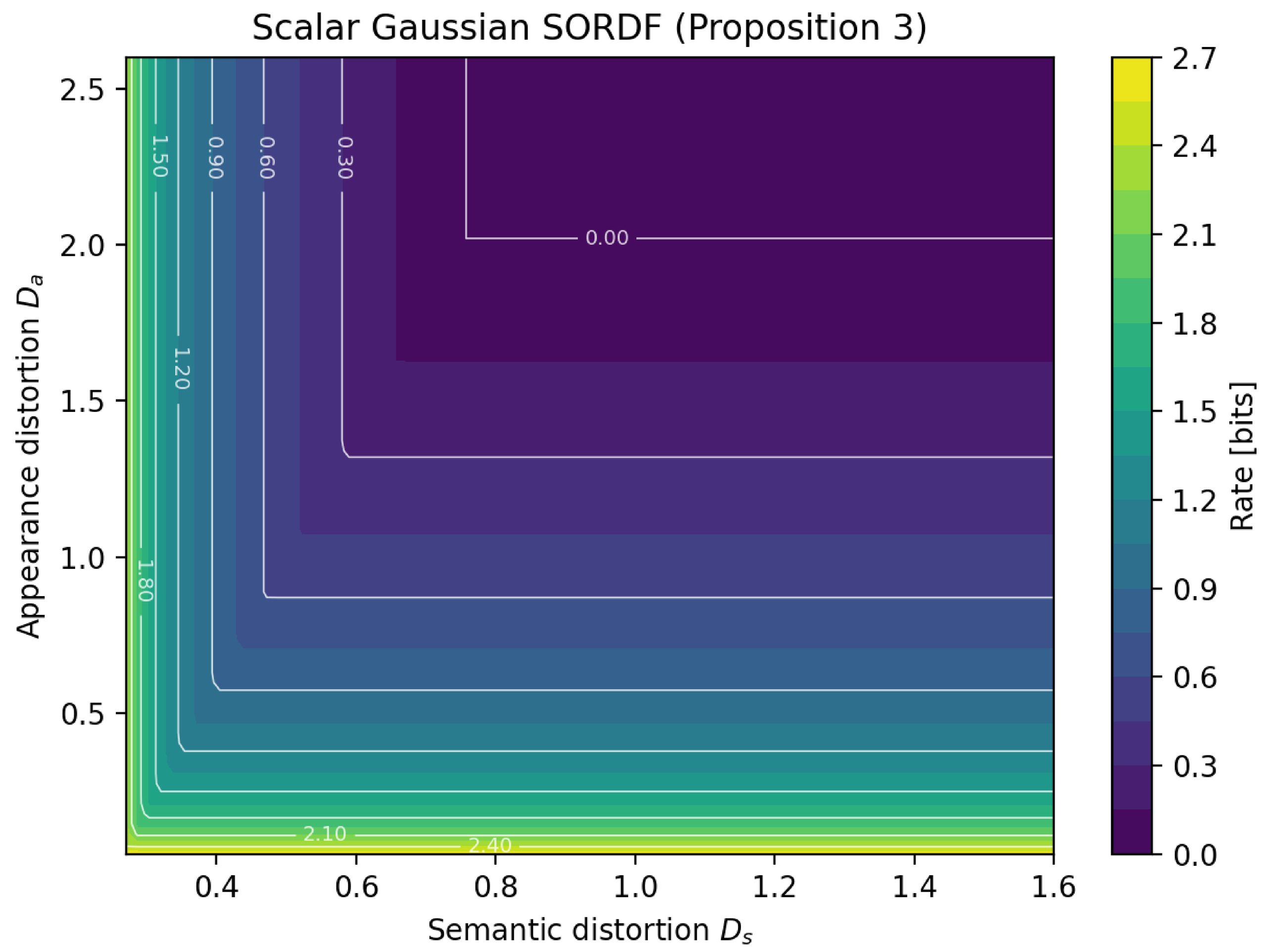
Proposition 5 computes $R(D_s, \infty)$ by solving a Lagrange multiplier equation and evaluating an entropy integral over the Gaussian mixture.

Adding appearance fidelity increases the required rate



Proposition 6 adds an appearance-refinement term to the semantic rate. Tight D_a constraints require extra bits even when the semantic task is already satisfied.

Scalar Gaussian sources show an aligned two-constraint maximum

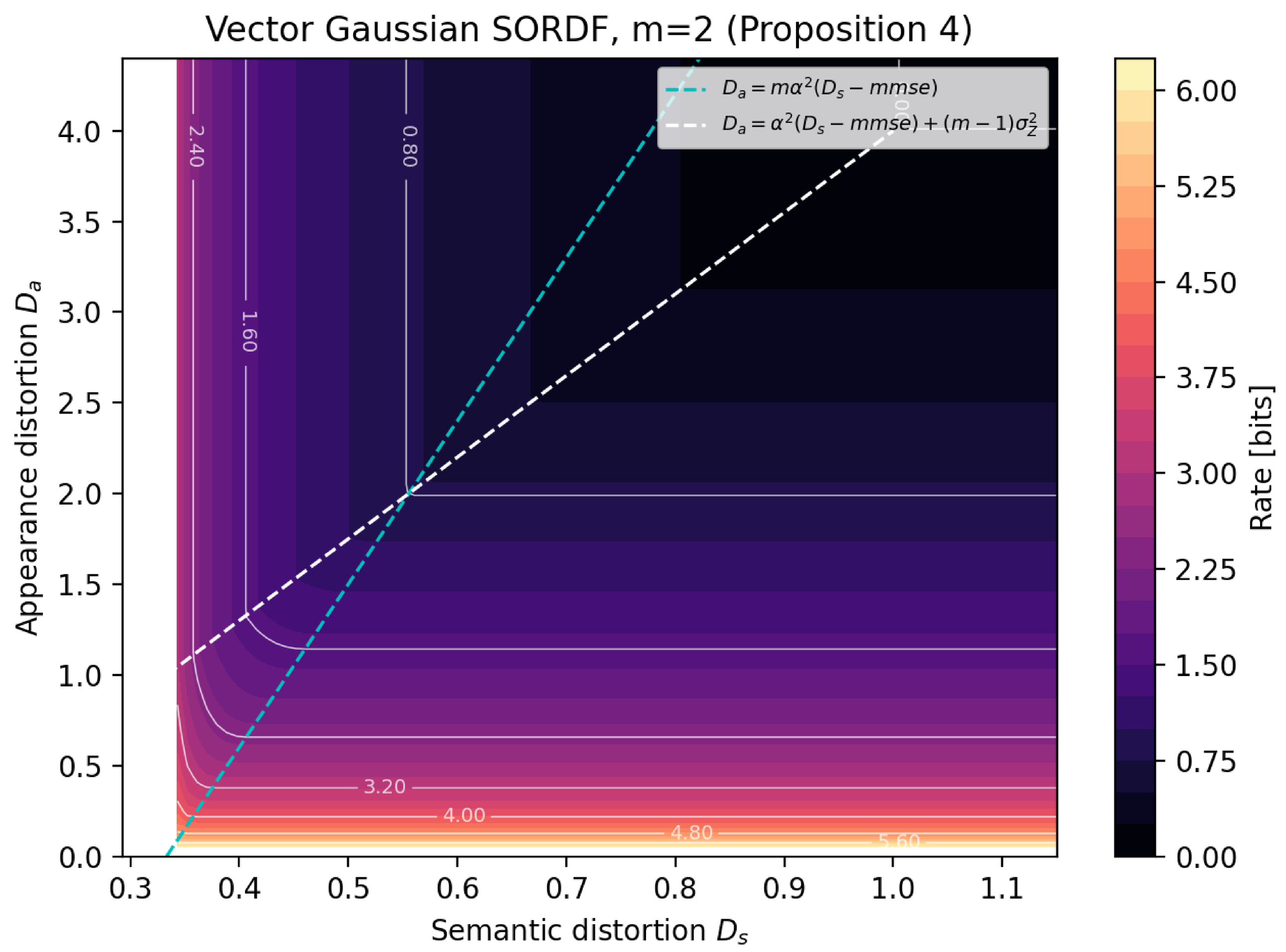


For scalar jointly Gaussian S and X, Proposition 3 gives

$$R = \frac{1}{2} \max\left\{\log^+ \frac{\Sigma_X}{D_a},\ \log^+ \frac{\Sigma_{SX}^2}{\Sigma_X (D_s - \mathrm{mmse})}\right\}.$$

The larger of the appearance and semantic requirements determines the rate.

Vector Gaussian sources reveal a genuine semantic–appearance tradeoff region



For $X = \mathbf{1}_m S + Z$, Proposition 4 divides the (D_s, D_a) plane into five regions. The slanted boundaries mark where semantic and appearance constraints interact.

Takeaways

- Semantic information can be studied using rate–distortion tools.
- A hidden state S makes semantic reconstruction an indirect coding problem.
- Binary and Gaussian cases expose different tradeoff geometry.
- The reproduced figures provide report-ready evidence for the framework.

Source paper: Jiakun Liu, Wenyi Zhang, and H. Vincent Poor, "A Rate-Distortion Framework for Characterizing Semantic Information," 2021. Code and generated figures are version controlled in `rd-semantic-repro`.